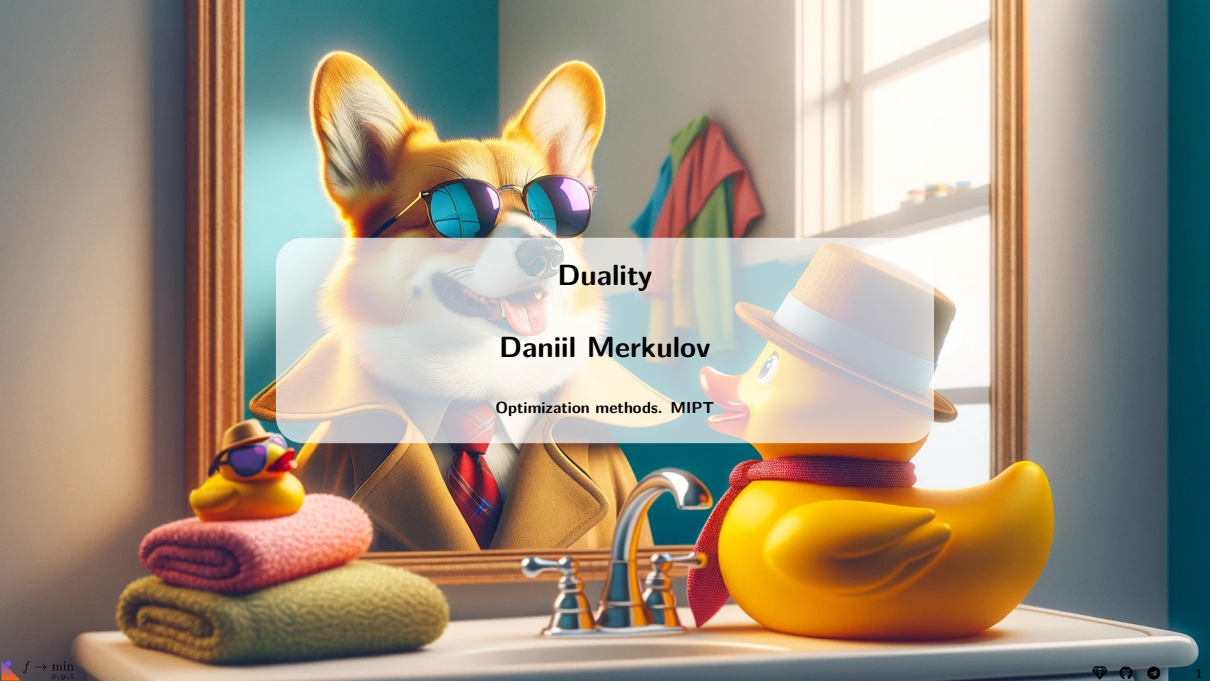


Duality

Daniil Merkulov

Optimization methods. MIPT

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Motivation

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As a consequence:

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And the Lagrangian, associated with this problem:

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x) = f_0(x) + \lambda^\top f(x) + \nu^\top h(x)$$

Dual function

We assume $\mathcal{D} = \bigcap_{i=0}^{m} \mathbf{dom}\ f_i \cap \bigcap_{i=1}^{p} \mathbf{dom}\ h_i$ is nonempty. We define the Lagrange dual function (or just dual function) $g : \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R}$ as the minimum value of the Lagrangian over x: for $\lambda \in \mathbb{R}^m, \nu \in \mathbb{R}^p$

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When the Lagrangian is unbounded below in x, the dual function takes on the value $-\infty$. Since the dual function is the pointwise infimum of a family of affine functions of (λ, ν), it is concave, even when the original problem is not convex.

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Let us show, that the dual function yields lower bounds on the optimal value p^* of the original problem for any $\lambda \succeq 0, \nu$. Suppose some $\hat{x}$ is a feasible point for the original problem, i.e., $f_i(\hat{x}) \leq 0$ and $h_i(\hat{x}) = 0,\ \lambda \succeq 0$. Then we have:

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The term "dual feasible", to describe a pair (λ, ν) with $\lambda \succeq 0$ and $g(\lambda, \nu) > -\infty$, now makes sense. It means, as the name implies, that (λ, ν) is feasible for the dual problem. We refer to (λ^*, ν^*) as dual optimal or optimal Lagrange multipliers if they are optimal for the above problem.

Summary

	Primal	Dual
Function	$f_0(x)$	$g(\lambda, \nu) = \min\limits_{x \in \mathcal{D}} L(x, \lambda, \nu)$
Variables	$x \in S \subseteq \mathbb{R}^n$	$\lambda \in \mathbb{R}^m_+, \nu \in \mathbb{R}^p$
Constraints	$f_i(x) \leq 0,\ i = 1, \ldots, m$ $h_i(x) = 0,\ i = 1, \ldots, p$	$\lambda_i \geq 0, \forall i \in \overline{1, m}$
Problem	$f_0(x) \to \min\limits_{x \in \mathbb{R}^n}$ s.t. $f_i(x) \leq 0,\ i = 1, \ldots, m$ $h_i(x) = 0,\ i = 1, \ldots, p$	$g(\lambda, \nu) \to \max\limits_{\lambda \in \mathbb{R}^m, \nu \in \mathbb{R}^p}$ s.t. $\lambda \succeq 0$
Optimal	x^* if feasible, $p^* = f_0(x^*)$	λ^*, ν^* if $\max$ is achieved, $d^* = g(\lambda^*, \nu^*)$

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$$-(1/4)\nu^T A A^T \nu - b^T \nu \leq \inf\{x^T x \,|\, Ax = b\}.$$

Which is a simple non-trivial lower bound without any problem solving.

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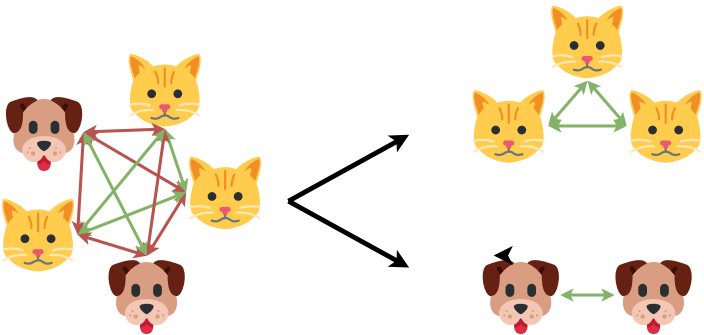


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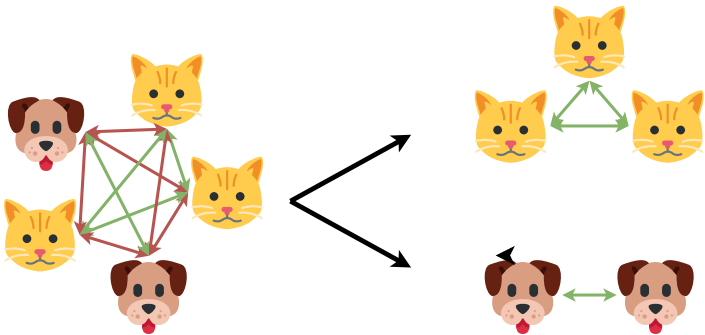


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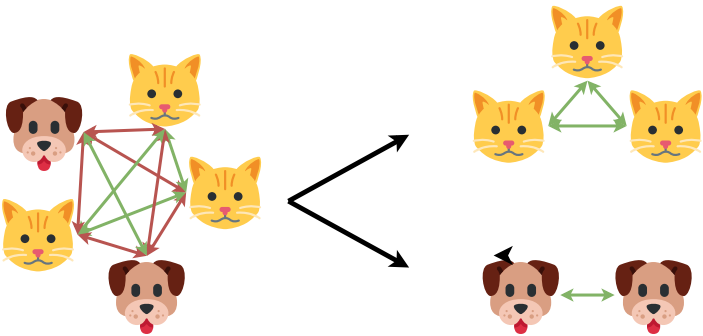


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The coefficient W_{ij} in the matrix represents the expense associated with placing elements i and j in the same partition, while $-W_{ij}$ signifies the cost of segregating them. The objective encapsulates the aggregate cost across all pairs of elements, and the challenge posed by problem is to find the partition that minimizes the total cost.

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We now derive the dual function for this problem. The Lagrangian is expressed as

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This renders a simple bound on the optimal value p^*: $p^* \geq -\mathbf{1}^T \nu = n\lambda_{\mathsf{min}}(W)$.

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$$g(\nu) = \inf_x x^T (W + \mathsf{diag}(\nu)) x - \mathbf{1}^T \nu = \begin{cases} -\mathbf{1}^T \nu & \text{if } W + \mathsf{diag}(\nu) \succeq 0 \\ -\infty & \text{otherwise,} \end{cases}$$

exploiting the realization that the infimum of a quadratic form is either zero (when the form is positive semidefinite) or $-\infty$ (when it's not).

This dual function furnishes lower bounds on the optimal value of the problem. For instance, we can adopt the particular value of the dual variable

$$\nu = -\lambda_{\mathsf{min}}(W)\mathbf{1}$$

which is dual feasible, since $W + \mathsf{diag}(\nu) = W - \lambda_{\mathsf{min}}(W) I \succeq 0$.

This renders a simple bound on the optimal value p^*: $p^* \geq -\mathbf{1}^T \nu = n\lambda_{\mathsf{min}}(W)$.

The code for the problem is available here Open in Colab

Strong duality

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- "Easy" necessary and sufficient conditions: unknown.

Strong duality in linear least squares

Exercise

In the Least-squares solution of linear equations example above calculate the primal optimum p^* and the dual optimum d^* and check whether this problem has strong duality or not.

Useful features of duality

- **Construction of lower bound on solution of the primal problem.**

  It could be very complicated to solve the initial problem. But if we have the dual problem, we can take an arbitrary $y \in \Omega$ and substitute it in $g(y)$ - we'll immediately obtain some lower bound.

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- **Dual function is always concave**

  As a pointwise minimum of affine functions.

Slater's condition

Theorem

If for a convex optimization problem (i.e., assuming minimization, f_0, f_i are convex and h_i are affine), there exists a point x such that $h(x) = 0$ and $f_i(x) < 0$ (existance of a strictly feasible point), then we have a zero duality gap and KKT conditions become necessary and sufficient.

An example of convex problem, when Slater's condition does not hold

Example

$$\min\{f_0(x) = x \mid f_1(x) = \frac{x^2}{2} \leq 0\},$$

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Example

$$\min\{f_0(x) = x \mid f_1(x) = \frac{x^2}{2} \leq 0\},$$

The only point in the budget set is: $x^* = 0$. However, it is impossible to find a non-negative $\lambda^* \geq 0$, such that

$$\nabla f_0(0) + \lambda^* \nabla f_1(0) = 1 + \lambda^* x = 0.$$

A nonconvex quadratic problem with strong duality

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where $A \in \mathbb{S}^n$, $A \not\succeq 0$ and $b \in \mathbb{R}^n$. Since $A \not\succeq 0$, this is not a convex problem. This problem is sometimes called the trust region problem, and arises in minimizing a second-order approximation of a function over the unit ball, which is the region in which the approximation is assumed to be approximately valid.

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$$-\sum_{i=1}^n \frac{(q_i^\top b)^2}{\lambda_i+\lambda} - \lambda \to \max_{\lambda\in\mathbb{R}}$$
$$\text{s.t. } \lambda \ge -\lambda_{min}(A)$$

Applications

Solving the primal via the dual

An important consequence of stationarity: under strong duality, given a dual solution λ^*, ν^*, any primal solution x^* solves

$$\min_{x \in \mathbb{R}^n} f_0(x) + \sum_{i=1}^{m} \lambda_i^* f_i(x) + \sum_{i=1}^{p} \nu_i^* h_i(x)$$

Often, solutions of this unconstrained problem can be expressed **explicitly**, giving an explicit characterization of primal solutions from dual solutions.

Furthermore, suppose the solution of this problem is unique; then it must be the primal solution x^*.

This can be very helpful when the dual is easier to solve than the primal.

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This gives:

$$x_i^\star = \frac{a_i \nu^\star}{c_i}.$$

Mixed strategies for matrix games



Player 1

u_1
$\dots$
u_k
$\dots$
u_n

v_1
$\dots$
$\dots$
v_l
$\dots$
$\dots$
v_m



Player 2

Figure 2: The scheme of a mixed strategy matrix game

Mixed strategies for matrix games



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$\dots$

u_k

$\dots$

u_n

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In zero-sum matrix games, players 1 and 2 choose actions from sets $\{1, ..., n\}$ and $\{1, ..., m\}$, respectively. The outcome is a payment from player 1 to player 2, determined by a payoff matrix $P \in \mathbb{R}^{n \times m}$. Each player aims to use mixed strategies, choosing actions according to a probability distribution: player 1 uses probabilities u_k for each action i, and player 2 uses v_l.

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In zero-sum matrix games, players 1 and 2 choose actions from sets $\{1, ..., n\}$ and $\{1, ..., m\}$, respectively. The outcome is a payment from player 1 to player 2, determined by a payoff matrix $P \in \mathbb{R}^{n \times m}$. Each player aims to use mixed strategies, choosing actions according to a probability distribution: player 1 uses probabilities u_k for each action i, and player 2 uses v_l.

The expected payoff from player 1 to player 2 is given by $\sum_{k=1}^{n} \sum_{l=1}^{m} u_k v_l P_{kl} = u^T P v$. Player 1 seeks to minimize this expected payoff, while player 2 aims to maximize it.

Figure 2: The scheme of a mixed strategy matrix game

Mixed strategies for matrix games. Player 1's Perspective



Player 1

u_1
$\dots$
u_k
$\dots$
u_n

Assuming player 2 knows player 1's strategy u, player 2 will choose v to maximize $u^T P v$. The worst-case expected payoff is thus:

$$\max_{v \geq 0, 1^T v = 1} u^T P v = \max_{i=1,\dots,m} (P^T u)_i$$

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Player 1's optimal strategy minimizes this worst-case payoff, leading to the optimization problem:

$$\begin{aligned} &\min \max_{i=1,\dots,m} (P^T u)_i \\ &\text{s.t. } u \geq 0 \\ &1^T u = 1 \end{aligned} \tag{1}$$

This forms a convex optimization problem with the optimal value denoted as p_1^*.

Mixed strategies for matrix games. Player 2's Perspective

v_1
$\dots$
$\dots$
v_l
$\dots$
$\dots$
v_m



Player 2

Conversely, if player 1 knows player 2's strategy v, the goal is to minimize $u^T P v$. This leads to:

$$\min_{u \geq 0, 1^T u = 1} u^T P v = \min_{i=1,\dots,n} (Pv)_i$$

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$$\min_{u \geq 0, 1^T u = 1} u^T P v = \min_{i=1,\dots,n} (Pv)_i$$

Player 2 then maximizes this to get the largest guaranteed payoff, solving the optimization problem:

$$\begin{aligned} &\max \min_{i=1,\dots,n} (Pv)_i \\ &\text{s.t. } v \geq 0 \\ &1^T v = 1 \end{aligned} \tag{2}$$

The optimal value here is p_2^*.

Mixed strategies for matrix games

Duality and Equivalence

It's generally advantageous to know the opponent's strategy, but surprisingly, in mixed strategy matrix games, this advantage disappears. The key lies in duality: the problems above are Lagrange duals. By formulating player 1's problem as a linear program and introducing Lagrange multipliers, we find that the dual problem matches player 2's problem. Due to strong duality in feasible linear programs, $p_1^* = p_2^*$, showing no advantage in knowing the opponent's strategy.

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Conclusion

This formulation shows that the Lagrange dual problem is equivalent to problem Equation 2. Given the feasibility of these linear programs, strong duality holds, meaning the optimal values of Equation 1 and Equation 2 are equal.

References

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- Duality Uses and Correspondences lecture by Ryan Tibshirani course.